

Cost Function ~~Intuition~~ Intuition

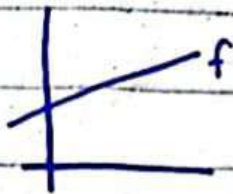
Now we will see how cost function can be used to find best params for your model.

model:

$$f_{w,b}(x) = wx + b \quad \text{--- (i)}$$

parameters:

w, b



cost function:

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

goal:

$$\text{minimize}_{w,b} J(w,b)$$

In order to better visualize cost function (J), we'll take simplified version of linear reg model:

$$f_w(x) = wx \rightarrow \text{model} \quad \left(\begin{array}{l} \text{just took out } b \\ \text{or } b=0 \text{ from (i)} \end{array} \right)$$

params:

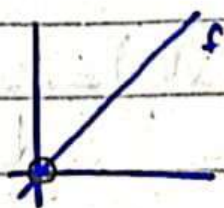
w

cost function:

$$J(w) = \frac{1}{2m} \sum_{i=1}^m (f_w(x^{(i)}) - y^{(i)})^2$$

goal:

$$\text{minimize}_w J(w)$$



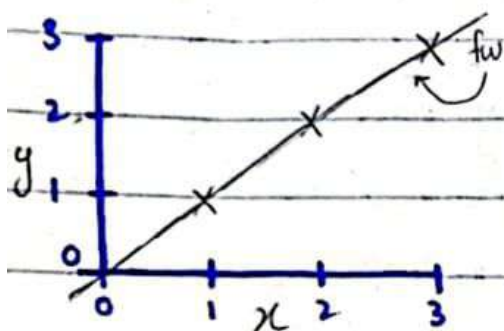
How two are related (graphically)

$w=1$

$f_w(x)$

(for fixed w , then f_w is)
only function of x)

means estimated value of y
depends on value of input x .



Say $w=1$, so for this choice
of 1, the function f_w looks
like this straight line with
slope of 1.

Now, calculate:

$J(w)$ when $w=1$

$$J(w) = \frac{1}{2m} \sum_{i=1}^m (f_w(x^{(i)}) - y^{(i)})^2$$

$$= \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} - y^{(i)})^2$$

for above dataset:

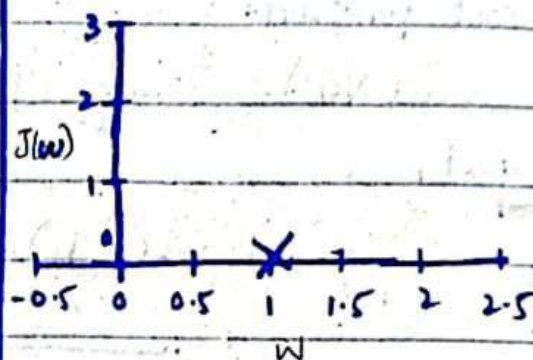
$$J(w) = 0 \text{ as when } x=1, y=1$$

$$\text{as } = \frac{1}{2m} (0^2 + 0^2 + 0^2) \quad x=2, y=2$$

$J(w)$

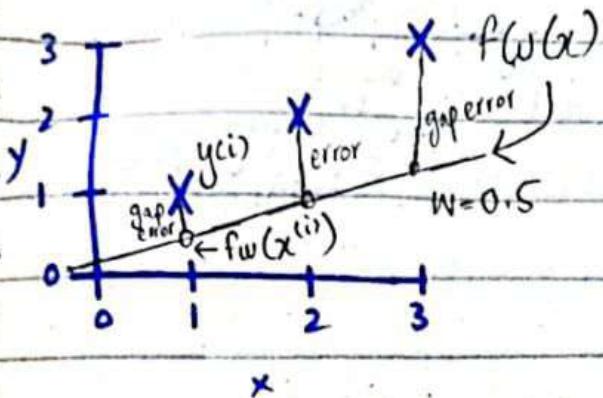
cost function J is function
of w , where w
controls the slope of line
defined by f_w .
So cost defined by J
depends on parameter

$\rightarrow w$

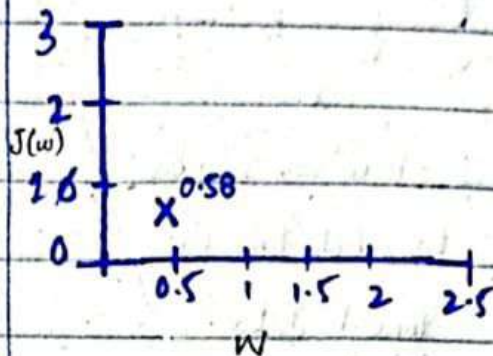


Same example with $w = 0.5$

$f_w(x)$
(func. of x)



$J(w)$
(func of w)



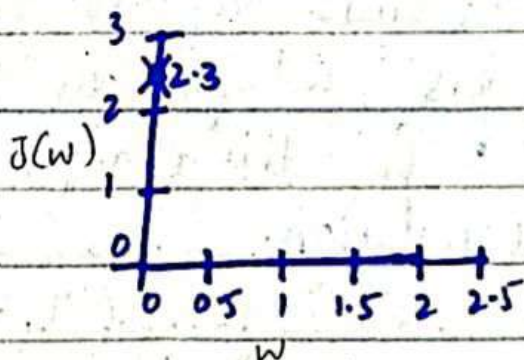
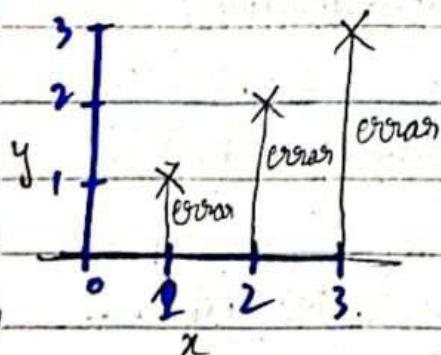
$$J(0.5) = \frac{1}{2m} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2]$$

$$= \frac{1}{2 \cdot 3} [3.5]$$

$$\text{as } m=3 \quad 2 \cdot (3)$$

$$= \frac{1}{6} (3.5) \approx \boxed{0.58}$$

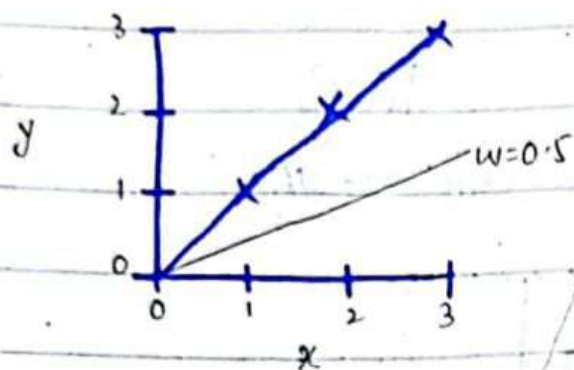
Now $w=0$



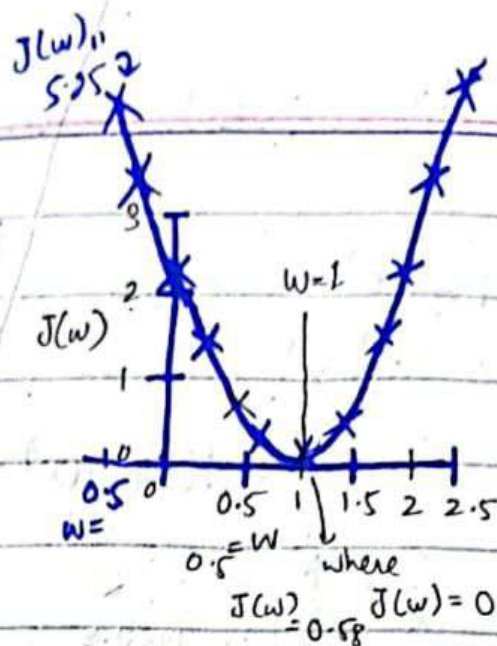
$$J(0) = \frac{1}{2m} [(0-1)^2 + (0-2)^2 + (0-3)^2]$$

$$= \frac{1}{6} (1 + 4 + 9) = \frac{14}{6}$$

$$= \boxed{2.3}$$



$w=1$ → here corresponds
to whole line fit
through data



- So for each value of w you end up with a different line and corresponding cost $J(w)$
- Given this how can you find/choose value of w that result in function f fitting data well.
- $J(w)$ is function that calculates how big the sq. of errors are so choosing w that minimize these sq. errors will give us good model.
- for this example $w=1$ (fits data best well)

Conclusion:

The goal of linear regression is to find params w & b that result in smallest possible value for c.f J .